

MATH3303 – Lecture 19

Rings, Subrings, Zero Divisors, Integral Domains, and Fields

Review Notes

Reference: Lee, Abstract Algebra, Sections 8.1–8.4

Overview

This lecture reviews the foundational structures in ring theory:

- **Rings** – the basic algebraic objects
- **Subrings** – substructures of rings
- **Zero divisors** – nonzero elements whose product is zero
- **Integral domains** – well-behaved commutative rings
- **Fields** – the “best” commutative rings (every nonzero element is invertible)

Historical note. Dedekind introduced the ring of integers of a number field in 1871.

1 Rings

Definition 1.1: Ring

A *ring* is a set R equipped with two binary operations, called *addition* $(+)$ and *multiplication* $(\cdot)$, such that:

(R1) $(R, +)$ is an abelian group;

(R2) $(R, \cdot)$ is a monoid (i.e., multiplication is associative): $(ab)c = a(bc)$;

(R3) Distributive laws hold:

$$a(b + c) = ab + ac \quad \text{and} \quad (b + c)a = ba + ca.$$

We write ab for $a \cdot b$.

Definition 1.2: Commutative & Unital Rings

A ring R is:

- *commutative* if $ab = ba$ for all $a, b \in R$;
- *unital* (or *a ring with identity*) if there exists $1 \in R$ such that $1 \cdot a = a \cdot 1 = a$ for all $a \in R$.

Remark 1.1: Convention

In this course we always assume rings are unital.

Example 1.1: Standard examples of rings (from lecture)

1. $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Z}_n$ are all commutative rings with identity.

2. $\mathbb{H} = \mathbb{R} \oplus \mathbb{R}i \oplus \mathbb{R}j \oplus \mathbb{R}k$ is the ring of *quaternions*, where

$$i^2 = j^2 = k^2 = ijk = -1.$$

This is a non-commutative ring with identity.

3. For any ring R , the *polynomial ring* $R[x]$ is a ring with coefficients in R .

4. For any ring R and $n \geq 2$, the set $M_n(R)$ of $n \times n$ matrices with entries in R is a ring.

Remark 1.2: Determinant identity

If we want $\det(AB) = \det(A)\det(B)$ to hold for all $A, B \in M_n(R)$, then R must be commutative.

Example 1.2: More examples of rings

1. **Polynomials with real coefficients:** $\mathbb{R}[x]$ is a commutative ring with identity. Similarly $\mathbb{Z}[x]$.

2. **Even integers:** $2\mathbb{Z}$ is a commutative ring *without* identity (no even integer multiplies 2 to give 2).
3. **Gaussian integers:** $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$ is a commutative ring with identity.
4. **Direct sum:** If R and S are rings, then $R \oplus S$ with componentwise operations is also a ring.

Things that can go wrong in rings

These “surprises” show how rings differ from familiar arithmetic in $\mathbb{R}$.

Example 1.3: $ab = 0$ does not imply $a = 0$ or $b = 0$

In $\mathbb{Z}_6$, we have $2 \cdot 3 = 0$, yet $2 \neq 0$ and $3 \neq 0$. In $M_n(\mathbb{R})$:

$$\begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix} \begin{pmatrix} -2 & 4 \\ 1 & -2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix},$$

yet neither factor is the zero matrix.

Example 1.4: Cancellation can fail

In $\mathbb{Z}_{12}$: $3 \cdot 1 = 3 \cdot 5 = 3$, but $1 \neq 5$.

Example 1.5: $ab = b$ does not imply $a = 1$

In $\mathbb{Z}_{12}$: $5 \cdot 3 = 3$, but $5 \neq 1$. To verify $a = 1$, we must check $ab = b = ba$ for *every* $b \in R$.

2 Subrings

Definition 2.1: Subring

Let R be a ring. A subset $S \subseteq R$ is a *subring* of R if:

- (1) $0 \in S$;
- (2) S is closed under subtraction: $a, b \in S \implies a - b \in S$;
- (3) S is closed under multiplication: $a, b \in S \implies ab \in S$.

Equivalently, S is an additive subgroup of R that is closed under multiplication.

Remark 2.1: Subrings need not contain 1

A subring need not contain the multiplicative identity of R . If it does, we call it a *unital subring*.

Example 2.1: Examples of subrings (from lecture)

1. $n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$ is a subring of $\mathbb{Z}$ for each $n \in \mathbb{Z}$. For $n \geq 2$, $n\mathbb{Z}$ is *not* a unital subring.
2. Upper triangular matrices form a subring of $M_n(R)$. Similarly diagonal matrices.
3. For any ring R , both $\{0\}$ and R are subrings.

Example 2.2: $2\mathbb{Z}$ as a subring of $\mathbb{Z}$

$0 \in 2\mathbb{Z}$. If $2a, 2b \in 2\mathbb{Z}$ then $2a - 2b = 2(a - b) \in 2\mathbb{Z}$ and $(2a)(2b) = 2(2ab) \in 2\mathbb{Z}$. So $2\mathbb{Z}$ is a subring (but not unital).

Example 2.3: A subring with its own identity, different from R 's

Let $R = \mathbb{Z}_6$ and $S = \{0, 3\}$. Check: $3 + 3 = 6 \equiv 0$, $3 \cdot 3 = 9 \equiv 3$. So S is a subring, and 3 acts as an identity *within* S (since $3 \cdot 0 = 0$, $3 \cdot 3 = 3$). However $3 \neq 1$ in $\mathbb{Z}_6$, so S is not a unital subring of R .

Example 2.4: The matrix subring of 2×2 matrices

The set $S = \left\{ \begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix} : a \in \mathbb{R} \right\}$ is a subring of $M_2(\mathbb{R})$. Closure under subtraction and multiplication follow from direct calculation. It has its own identity $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, which is not the identity of $M_2(\mathbb{R})$, so it is not a unital subring.

Center of a Ring

Definition 2.2: Center

Let R be a ring. The *center* of R is

$$Z(R) = \{r \in R : rs = sr \text{ for all } s \in R\}.$$

Theorem 2.1: Center is a unital subring

For any ring R , the center $Z(R)$ is a unital subring of R .

Proof. $0a = 0 = a0$ for all a , so $0 \in Z(R)$. Take $y, z \in Z(R)$. For any $a \in R$:

$$a(y - z) = ay - az = ya - za = (y - z)a, \quad ayz = yaz = yza,$$

so $y - z, yz \in Z(R)$. Also $1 \in Z(R)$ since $1a = a = a1$. □

Example 2.5: Center of $M_n(\mathbb{R})$ (from lecture)

Suppose $A = (a_{ij}) \in Z(M_n(\mathbb{R}))$. Let E_{ij} denote the matrix with 1 in position (i, j) and 0 elsewhere. Then:

- $AE_{kk} = E_{kk}A$ for each k forces A to be *diagonal*.
- $AE_{ij} = E_{ij}A$ for $i \neq j$ forces $a_{ii} = a_{jj}$.

Therefore

$$Z(M_n(\mathbb{R})) = \{\lambda I_n : \lambda \in \mathbb{R}\} \quad (\text{scalar matrices}).$$

Example 2.6: Center of a commutative ring

If R is commutative, then $Z(R) = R$.

3 Zero Divisors

Definition 3.1: Zero divisor

Let R be a commutative ring. An element $a \in R \setminus \{0\}$ is called a *zero divisor* if there exists $b \in R \setminus \{0\}$ such that

$$ab = 0.$$

Example 3.1: Zero divisor in $\mathbb{Z}_6$ (from lecture)

In $\mathbb{Z}_6$, we have $2 \cdot 3 = 0$. So 2 (and also 3) is a zero divisor in $\mathbb{Z}_6$.

Example 3.2: More zero divisors

- In $\mathbb{Z}_6$: the zero divisors are $\{2, 3, 4\}$. The element 5 is not a zero divisor (since $\gcd(5, 6) = 1$).
- In $\mathbb{Z}$: there are no zero divisors.
- In $\mathbb{Z}_p$ (p prime): there are no zero divisors.
- In $\mathbb{Z}_2 \oplus \mathbb{Z}_2$: $(1, 0) \cdot (0, 1) = (0, 0)$, so $(1, 0)$ and $(0, 1)$ are zero divisors.

Remark 3.1: General rule for $\mathbb{Z}_n$

In $\mathbb{Z}_n$, an element $a \neq 0$ is a zero divisor if and only if $\gcd(a, n) > 1$.

4 Integral Domains

Definition 4.1: Integral domain

An *integral domain* is a commutative ring R with identity $1 \neq 0$ that has no zero divisors.

Remark 4.1: Why $1 \neq 0$? (from lecture)

If $1 = 0$ in R , then for any $a \in R$,

$$a = 1 \cdot a = 0 \cdot a = 0,$$

so $R = \{0\}$. We exclude this trivial ring from being an integral domain.

Example 4.1: Integral domain examples

- $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ are all integral domains.
- $\mathbb{Z}_p$ is an integral domain when p is prime.
- $\mathbb{R}[x]$ is an integral domain. More generally, $R[x]$ is an integral domain whenever R is.
- $\mathbb{Z}[i]$ (Gaussian integers) is an integral domain.
- $\mathbb{Z}_6$ is *not* an integral domain (zero divisors).
- $2\mathbb{Z}$ is *not* an integral domain (no identity).
- $M_n(\mathbb{R})$ for $n \geq 2$ is *not* an integral domain (not commutative, has zero divisors).

Lemma 4.1: Cancellation in integral domains (from lecture)

Let R be an integral domain and let $b, c \in R$. If $a \in R \setminus \{0\}$ satisfies

$$ab = ac,$$

then $b = c$.

Proof. Suppose $ab = ac$. Then

$$ab - ac = 0 \implies a(b - c) = 0.$$

Since R has no zero divisors, either $a = 0$ or $b - c = 0$. By assumption $a \neq 0$, so $b = c$. $\square$

5 Fields

Definition 5.1: Field and division ring

- A *field* is a commutative ring with $1 \neq 0$ in which every nonzero element has a multiplicative inverse.
- A *division ring* is a ring with $1 \neq 0$ (not necessarily commutative) in which every nonzero element has a multiplicative inverse.

Example 5.1: Fields and division rings

- $\mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{Z}_p$ (for p prime) are fields.
- The quaternions $\mathbb{H}$ form a division ring but *not* a field (non-commutative).
- $\mathbb{Z}$ is an integral domain but *not* a field (only ± 1 are invertible).

Theorem 5.1: Every field is an integral domain (from lecture)

If F is a field, then F is an integral domain.

Proof. We must show that if $a, b \in F$ and $ab = 0$, then $a = 0$ or $b = 0$.

Suppose $ab = 0$. If $a = 0$, we are done. Otherwise, $a \neq 0$, so a^{-1} exists. Multiplying both sides by a^{-1} :

$$a^{-1}(ab) = a^{-1} \cdot 0 \implies (a^{-1}a)b = 0 \implies 1 \cdot b = 0 \implies b = 0.$$

□

6 Units

Definition 6.1: Unit (from lecture)

Let R be a (unital) ring. An element $a \in R$ is called a *unit* if there exists $s \in R$ such that

$$as = sa = 1.$$

The set of all units of R is denoted by $R^\times$ (or $U(R)$):

$$R^\times = \{r \in R : \exists s \in R \text{ with } rs = sr = 1\}.$$

Remark 6.1: Uniqueness of inverses (from lecture)

If $r \in R^\times$ with $rs = 1$, and additionally $s'r = 1$, then $s = s'$:

$$s = s \cdot 1 = s(rs') = (sr)s' = 1 \cdot s' = s'.$$

So the inverse of a unit is unique; we denote it by r^{-1} .

Theorem 6.1: Units form a group

For any unital ring R , the set $R^\times$ is a group under multiplication.

Example 6.1: Units in standard rings

- $\mathbb{Z}^\times = \{1, -1\}$.
- $\mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}$ (and similarly for $\mathbb{R}, \mathbb{C}$).
- $M_n(\mathbb{R})^\times = GL_n(\mathbb{R})$, the group of invertible matrices.
- In a field F : $F^\times = F \setminus \{0\}$.
- $\mathbb{Z}_n^\times = U(n) = \{k : \gcd(k, n) = 1\}$.

7 Summary and Hierarchy

Property	Ring	Comm. Ring	Int. Domain	Field
$(R, +)$ abelian group	✓	✓	✓	✓
Multiplication associative	✓	✓	✓	✓
Distributive laws	✓	✓	✓	✓
Has $1 \neq 0$	(assumed)	(assumed)	✓	✓
$ab = ba$	—	✓	✓	✓
No zero divisors	—	—	✓	✓
Every $a \neq 0$ is a unit	—	—	—	✓

Hierarchy:

Fields $\subsetneq$ Integral Domains $\subsetneq$ Commutative Rings (with 1) $\subsetneq$ Rings (with 1).

8 Additional Key Theorems (Lee 8.3–8.4)

Theorem 8.1: Subring test (Lee 8.5)

Let R be a ring and $S \subseteq R$. Then S is a subring iff

1. $0 \in S$ (equivalently, $S \neq \emptyset$);
2. $a, b \in S \implies a - b \in S$;
3. $a, b \in S \implies ab \in S$.

Theorem 8.2: A unit cannot be a zero divisor (Lee Lemma 8.1)

In a commutative ring with identity, a unit cannot be a zero divisor.

Theorem 8.3: Finite integral domain is a field (Lee 8.10)

Every finite integral domain is a field.

Theorem 8.4: Classification of $\mathbb{Z}_n$ (Lee 8.11)

For $n \geq 2$, the following are equivalent:

1. $\mathbb{Z}_n$ is an integral domain;
2. $\mathbb{Z}_n$ is a field;
3. n is prime.

Theorem 8.5: Subfield test (Lee 8.12)

Let F be a field. Then $S \subseteq F$ is a subfield iff

1. $1 \in S$;
2. $a, b \in S \implies a - b \in S$;
3. $a, b \in S$ with $b \neq 0 \implies ab^{-1} \in S$.

9 Worked Example: $\mathbb{Q}(\sqrt{2})$ is a Subfield of $\mathbb{R}$

Let $F = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$.

1. $1 \in F$: $1 = 1 + 0\sqrt{2}$. ✓

2. **Closed under subtraction:**

$$(a_1 + b_1\sqrt{2}) - (a_2 + b_2\sqrt{2}) = (a_1 - a_2) + (b_1 - b_2)\sqrt{2} \in F.$$

3. **Closed under multiplication:**

$$(a_1 + b_1\sqrt{2})(a_2 + b_2\sqrt{2}) = (a_1a_2 + 2b_1b_2) + (a_1b_2 + a_2b_1)\sqrt{2} \in F.$$

4. **Inverses:** For $0 \neq a + b\sqrt{2} \in F$,

$$(a + b\sqrt{2})(a - b\sqrt{2}) = a^2 - 2b^2 \in \mathbb{Q}.$$

Since $\sqrt{2} \notin \mathbb{Q}$, we have $a^2 - 2b^2 \neq 0$. Therefore

$$(a + b\sqrt{2})^{-1} = \frac{a}{a^2 - 2b^2} - \frac{b}{a^2 - 2b^2}\sqrt{2} \in F.$$

10 Practice Problems

These are adapted from Lee, exercises 8.11–8.36.

Basic Properties of Rings

Problem 10.1: Expand

Let a, b be elements of a ring R . Simplify as far as possible:

1. $(a + b)(a - b)$;
2. $(a - b)^3$.

Hint: You cannot assume commutativity.

Problem 10.2: One-sided inverse implies zero

Let R be a ring with identity. Suppose $a, b, c \in R$ with $ab = ba = 1$ and $ac = 0$. Show that $c = 0$.

Problem 10.3: Left and right inverses

Let R be a ring with identity. Suppose $a, b, c \in R$ with $ba = ac = 1$. Does it follow that $b = c$? Prove or give a counterexample.

Problem 10.4: Idempotent ring

Let R be a ring in which $a^2 = a$ for every $a \in R$. Show that:

1. $a + a = 0$ for every $a \in R$;
2. R is commutative.

Problem 10.5: Nilpotent elements

Let R be a ring and $n > 2$ an integer. Show that if there exists $a \neq 0$ in R with $a^n = 0$, then there exists $b \neq 0$ in R with $b^2 = 0$.

Subrings**Problem 10.6: Gaussian integers are a subring**

Let $R = \{a + bi : a, b \in \mathbb{Z}\}$. Show R is a subring of $\mathbb{C}$. Is it a ring with identity? Is it a unital subring?

Problem 10.7: Strictly upper triangular 3×3 matrices

Let $R = \left\{ \begin{pmatrix} 0 & a & b \\ 0 & 0 & c \\ 0 & 0 & 0 \end{pmatrix} : a, b, c \in \mathbb{R} \right\}$. Show R is a subring of $M_3(\mathbb{R})$. Does R have an identity?

Problem 10.8: Diagonal-only matrices

Let R be the set of matrices $\begin{pmatrix} 0 & 0 \\ 0 & a \end{pmatrix}$ for $a \in \mathbb{R}$. Show R is a subring of $M_2(\mathbb{R})$. Does it have an identity? Is it a unital subring?

Problem 10.9: Intersection of subrings

Let R be a ring with subrings S and T . Show that $S \cap T$ is a subring. Generalize to any collection of subrings.

Problem 10.10: Annihilator

Let R be a ring and $a \in R$. Define $S = \{r \in R : ra = 0\}$. Is S necessarily a subring of R ? Prove it, or give a counterexample.

Problem 10.11: Principal right ideal

Let R be a ring and $a \in R$. Show that $S = \{ra : r \in R\}$ is a subring.

Problem 10.12: Center of $M_2(\mathbb{R})$

Show directly that the center of $M_2(\mathbb{R})$ is $\left\{ \begin{pmatrix} r & 0 \\ 0 & r \end{pmatrix} : r \in \mathbb{R} \right\}$.

Integral Domains, Fields, Zero Divisors and Units**Problem 10.13: Gaussian rationals form a subfield**

Let $R = \{a + bi : a, b \in \mathbb{Q}\}$. Show that R is a subfield of $\mathbb{C}$.

Problem 10.14: Units and zero divisors

For each of the following rings, identify which elements are units, and which are zero divisors:

1. $\mathbb{Z}_{18}$;
2. $\mathbb{Z}_3 \oplus \mathbb{Z}_9$.

Problem 10.15: Units of direct sums

Let R, S be rings with identity. Show that $U(R \oplus S) = U(R) \times U(S)$.

Problem 10.16: Idempotents in an integral domain

Show that every integral domain contains exactly two elements a satisfying $a^2 = a$.

Problem 10.17: When is $R \oplus S$ an integral domain?

Let R, S be rings. Under precisely what conditions is $R \oplus S$ an integral domain?

Problem 10.18: Intersection of subfields

Let F be a field with subfields K and L . Show that $K \cap L$ is a subfield. Generalize to arbitrary intersections.

Problem 10.19: Strong cancellation in an integral domain

Let R be an integral domain, and let $a, b \in R$ with $a^{13} = b^{13}$ and $a^{10} = b^{10}$. Show that $a = b$.

Problem 10.20: Finite commutative ring with no zero divisors

Let R be a finite commutative ring with no zero divisors. Show that either $R = \{0\}$ or R is an integral domain.

Solutions and Hints to Selected Problems

- **Expand:** (1) $a^2 + ba - ab - b^2$ (note: not $a^2 - b^2$!). (2) $a^3 - a^2b - aba - ba^2 + ab^2 + bab + b^2a - b^3$.
- **One-sided inverse implies zero:** $c = 1 \cdot c = (ba)c = b(ac) = b \cdot 0 = 0$.
- **Left and right inverses:** Yes. $b = b \cdot 1 = b(ac) = (ba)c = 1 \cdot c = c$.
- **Idempotent ring:** (1) Compute $(a+a)^2 = a+a$ and expand. (2) Compute $(a+b)^2 = a+b$ and use (1).
- **Nilpotent elements:** If $a^n = 0$ with n minimal, then $a^{n-1} \neq 0$. If n is even, say $n = 2k$, take $b = a^k$. If n is odd, take $b = a^{(n-1)/2}$ or adapt the argument.
- **Gaussian integers are a subring:** Verify the subring test. Identity exists $(1 + 0i)$. It is a unital subring of $\mathbb{C}$.
- **Strictly upper triangular 3×3 matrices:** Closure follows by matrix multiplication; the product of any two such matrices is again strictly upper triangular. It has no identity (you cannot get a 1 on the diagonal).
- **Diagonal-only matrices:** Subring check is straightforward. Identity is $\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$, but this is not I_2 . So it has identity but is not a unital subring.
- **Intersection of subrings:** $0 \in S \cap T$. If $a, b \in S \cap T$, then $a - b \in S$ and $a - b \in T$, so $a - b \in S \cap T$. Similarly for ab .
- **Annihilator:** Yes $-0 \in S$; if $r_1a = r_2a = 0$ then $(r_1 - r_2)a = 0$; and $(r_1r_2)a = r_1(r_2a) = 0$.
- **Principal right ideal:** $0 = 0 \cdot a$; $r_1a - r_2a = (r_1 - r_2)a$; $(r_1a)(r_2a) = (r_1ar_2)a$.
- **Center of $M_2(\mathbb{R})$:** Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in Z(M_2(\mathbb{R}))$. Compute $AE_{11} = E_{11}A$ to get $b = c = 0$; then $AE_{12} = E_{12}A$ gives $a = d$.
- **Gaussian rationals form a subfield:** Verify the subfield test. For $c+di \neq 0$, $(c+di)^{-1} = \frac{c}{c^2+d^2} - \frac{d}{c^2+d^2}i \in R$.
- **Units and zero divisors:**
 - In $\mathbb{Z}_{18}$: units are $\{1, 5, 7, 11, 13, 17\}$ (gcd with 18 equals 1); zero divisors are $\{2, 3, 4, 6, 8, 9, 10, 12, 14, 15, 16\}$.
 - In $\mathbb{Z}_3 \oplus \mathbb{Z}_9$: units are pairs (a, b) with $a \in U(\mathbb{Z}_3) = \{1, 2\}$ and $b \in U(\mathbb{Z}_9) = \{1, 2, 4, 5, 7, 8\}$. Everything else nonzero is a zero divisor.
- **Units of direct sums:** (r, s) has inverse (r_1, s_1) iff $rr_1 = 1$ and $ss_1 = 1$, i.e., $r \in U(R)$ and $s \in U(S)$.
- **Idempotents in an integral domain:** If $a^2 = a$, then $a(a - 1) = 0$. No zero divisors $\Rightarrow a = 0$ or $a = 1$. Both occur, giving exactly two.
- **When is $R \oplus S$ an integral domain?** $R \oplus S$ is an integral domain iff one of R, S is $\{0\}$ and the other is an integral domain. (Otherwise $(1, 0)(0, 1) = (0, 0)$ produces zero divisors.)
- **Intersection of subfields:** $K \cap L$ is a subring; $1 \in K \cap L$; inverses of nonzero elements lie in $K \cap L$ since both K and L contain them.

- **Strong cancellation:** From $a^{13} = b^{13}$ and $a^{10} = b^{10}$, deduce $a^3 = b^3$ (cancel a^{10} from $a^{13} = a^{10} \cdot a^3$ and $b^{13} = b^{10} \cdot b^3 = a^{10} \cdot b^3$). Combine $a^{10} = b^{10}$ with $a^3 = b^3$: write $\gcd(10, 3) = 1 = 10 \cdot 1 + 3 \cdot (-3)$, then $a = a^{10-9} = a^{10}(a^3)^{-3} \dots$ (work it out using cancellation).
- **Finite commutative ring with no zero divisors:** The only thing missing from being an integral domain is the identity. Use finiteness: for nonzero a , the map $r \mapsto ar$ is injective (by cancellation), hence surjective, so some e satisfies $ae = a$. Then check e acts as identity for all elements.

Quick Review Checklist

- ☐ State the definition of a ring (all three axioms).
- ☐ Distinguish rings, commutative rings, and unital rings.
- ☐ State and apply the subring test.
- ☐ Compute the center of $M_n(\mathbb{R})$.
- ☐ Identify zero divisors and units in $\mathbb{Z}_n$.
- ☐ State the definition of an integral domain and a field.
- ☐ Prove the cancellation law in an integral domain.
- ☐ Prove that every field is an integral domain.
- ☐ Use the subfield test on $\mathbb{Q}(\sqrt{2})$ or $\mathbb{Q}(i)$.
- ☐ Know that every finite integral domain is a field.
- ☐ Know that $\mathbb{Z}_n$ is a field iff n is prime.